

ARTIFICIAL INTELLIGENCE TRAINING COURSE

Artificial Brain: Neural Networks Explained Simply

Session 3 | Artificial Brain: Neural Networks Explained Simply

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TODAY'S ROADMAP

Previous Session Recap & Today's Objectives

Session 3 Objectives

Nature-Inspired: Understanding human brain neuron functions

Network Architecture: Understanding layers & decision weights

Deep Learning: The secret power behind multi-layer networks

Everyday Applications: Face recognition, translation & synthesis

Session 2 Recap

Machine Learning Logic Made Simple:

- Traditional programming vs. learning from data
- Three core pillars (data, model, prediction)
- Three types of learning (supervised, unsupervised, reinforcement)
- Hands-on workshop: building a model without coding

NATURE-INSPIRED

How Does the Human Brain Work?



86 Billion

Active neurons in the human brain

Signal Input: Receiving signals from the five senses

Group Processing: Rapid electrical signal transfer between neurons

Pathway Strengthening: Stronger connections through repetition & practice

DIRECT COMPARISON

What is an Artificial Neural Network?

Natural Human Brain

Building Block: Biological nerve cells

Connecting Bridge: Biological synapses

Learning Method: Daily experience & repetition

Artificial Neural Network

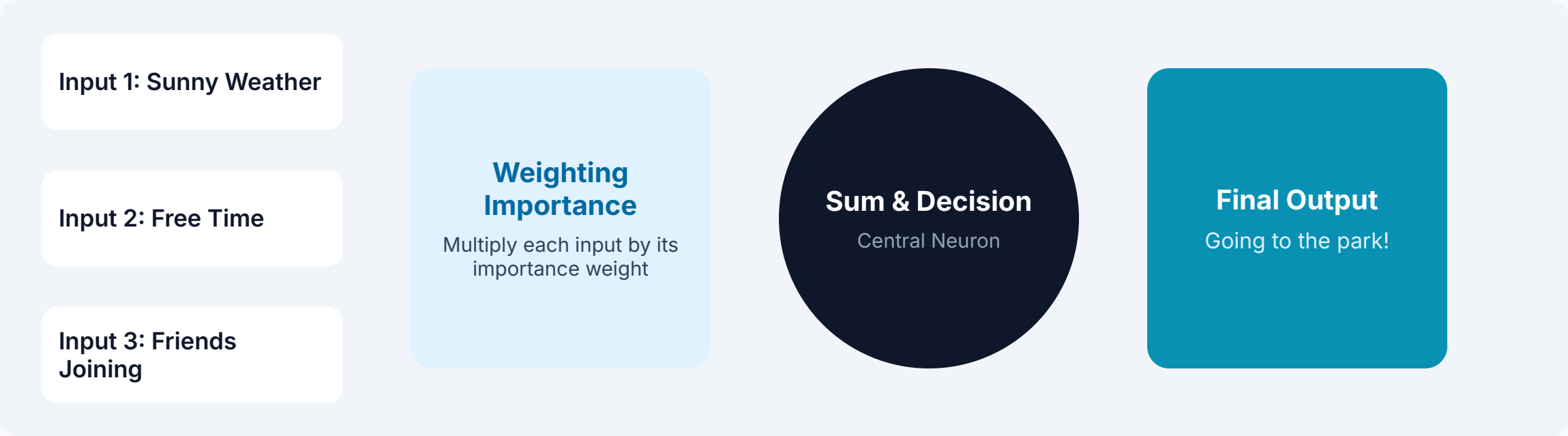
Building Block: Software computational nodes

Connecting Bridge: Adjustable numerical weights

Learning Method: Observing thousands of data samples

ANATOMY OF A NODE

Artificial Neuron Made Simple



1. Inputs

Raw initial data points fed into the neuron.

2. Weights

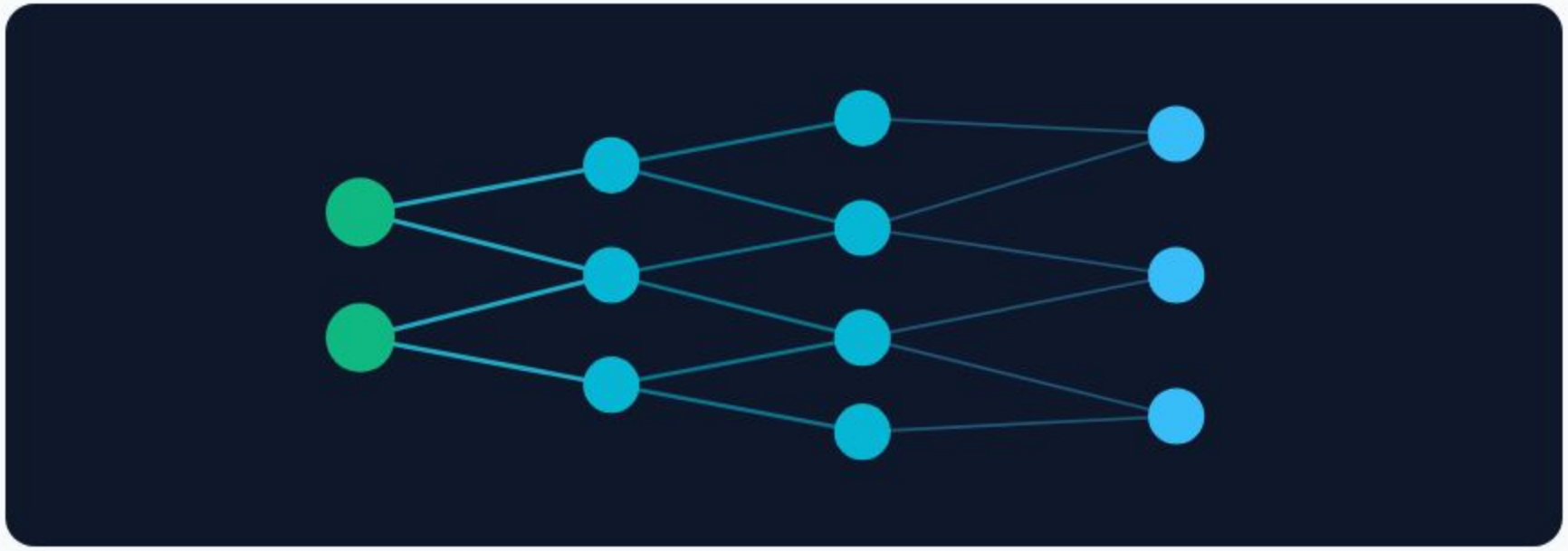
The degree of importance assigned to each input feature.

3. Output

The final result, response, or decision reached.

NETWORK ARCHITECTURE

Layers in Neural Networks



1. Input Layer

Receives image, text, or audio

2. Hidden Layers

Extracts patterns & hidden features

3. Output Layer

Delivers final diagnosis result

TRAINING & REFINEMENT CYCLE

Weights & Learning: How Does the Network Get Smart?



1. Initial Guess

Starting with completely random weights



2. Error Measurement

Comparing network guess with correct answer



3. Weight Adjustment

Tuning weights to reduce error



4. Full Mastery

High accuracy after thousands of iterations

Concrete Example: Just like adjusting shower water temperature by gradually turning the faucet!

CONCEPT OF DEPTH IN NETWORKS

What is Deep Learning?

Step-by-step learning from simple to complex:

Early Layers: Detecting dots, edges, and simple lines

Middle Layers: Combining lines to recognize shapes & components

Deep Final Layers: Full understanding of overall image or text context

Dozens of Layers

Sequential hidden layers working together

MODERN AI REVOLUTION

Why is Deep Learning so Powerful?



GPU Processing Power

Calculating millions of weights in seconds



Scales with Big Data

Gets smarter as more data is provided

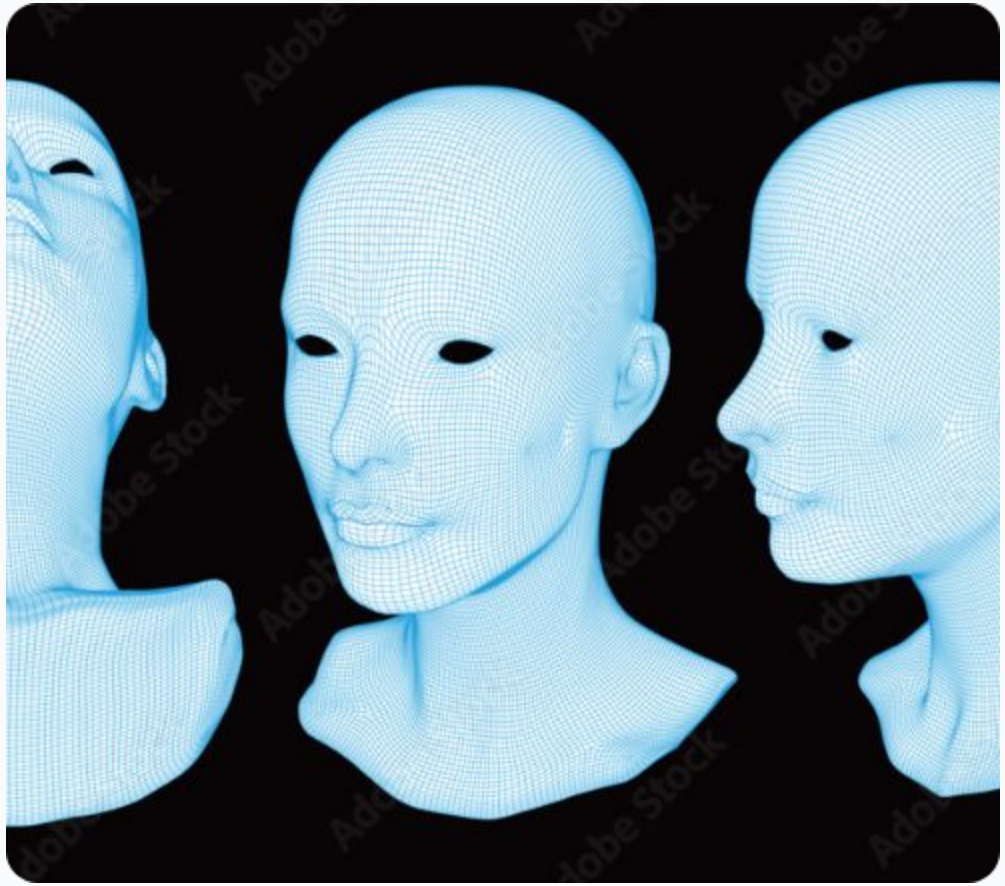


Automatic Feature Extraction

No need for manual rule programming

APPLICATION 1: COMPUTER VISION

Real-World Example: Facial Recognition on Smartphones



First Layer (Lines): Detecting light boundaries and image edges

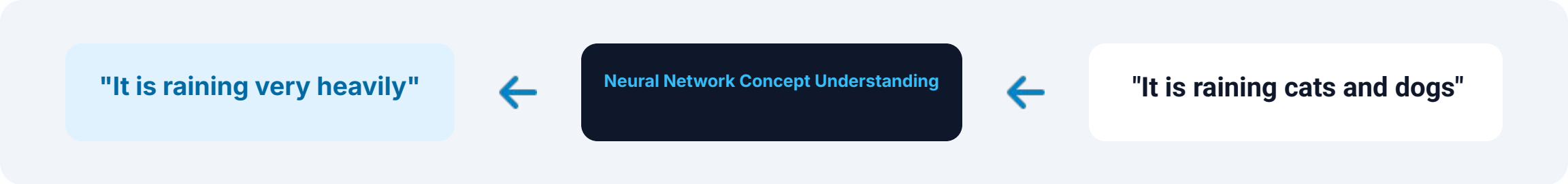
Second Layer (Features): Identifying eyes, eyebrows, nose, and lips

Third Layer (Identity): Matching full facial structure to the user

High Robustness: Accurate detection even with glasses or lighting changes

APPLICATION 2: NATURAL LANGUAGE PROCESSING

Real-World Example: Smart Language Translation



Beyond Word-for-Word Translation: Understanding full sentence meaning instead of dictionary translation

Context Awareness: Detecting idioms, metaphors, and tone of the author

Fluent Generation: Rewriting text naturally in the target language

APPLICATION 3: GENERATIVE AI

Real-World Example: Text-to-Image Generation



Text Prompt: "An astronaut cat in a starry galaxy"

Visual Imagination: Combining learned concepts from millions of artwork images

Denoising to Image: Creating clear images from random noise points

Novel Creation: Generating images that never existed before

BEFORE & AFTER COMPARISON

Traditional Methods vs. Deep Learning

Application Area	Traditional Programming (Before)	Deep Neural Networks (After)
Machine Translation	Rigid and word-for-word translation	**Fluent and conceptual translation**
Facial Recognition	Errors with glasses or angle changes	**Instant recognition in any lighting**
Speech Recognition	Requires speaking slowly and clearly	**Accurate understanding of casual speech**
Image Generation	Impossible without a human designer	**Artwork generation in seconds**

IT'S YOUR TURN!

Interactive Class Activity & Discussion



“So far today, which phone apps have you used that rely on neural networks?”

Hint: Face unlock, Instagram recommendations, navigation, smart keyboard, or camera filters

REALISTIC PERSPECTIVE

What Are the Limitations of Neural Networks?



High Data Demand

Requires thousands of samples to learn



Black Box Dilemma

Difficulty explaining exact decision logic



Energy & Cost Demand

Needs expensive servers and hardware

FINAL RECAP

Summary: What Did We Learn Today?

1. Brain Inspiration: Neural networks are designed inspired by human brain neurons.

2. Three-Part Structure: Each network consists of an input layer, hidden layers, and an output layer.

3. Learning with Weights: Learning means gradually adjusting weights to reduce prediction error.

4. Power of Deep Learning: Increasing hidden layers enables understanding complex patterns.

5. Everyday Impact: Face recognition, smart translation, and image generation are products of this technology.

FAQ & CLARIFICATIONS

Frequently Asked Questions About Neural Networks

Do Neural Networks Have Feelings?

No, it is simply a mathematical system for discovering statistical patterns.

Does It Require Heavy Mathematics?

No, today's pre-built tools have made the process simple for everyone.

Difference Between ML & Deep Learning?

Deep learning is an advanced branch of machine learning using multi-layer networks.

PRACTICAL & OBSERVATIONAL EXERCISE

Session 3 Assignment

Translation Test: Try a Persian proverb in Google Translate or chatbots.

Visual Search: Search for "sunset" or "car" in your phone gallery.

Record Observation: Note one instance of high accuracy and one of AI error.



Neural Network Hunter!

Test two smart tools during the week

End of Session 3 | Introduction to AI Course

Thank You for Your Participation & Attendance!

Next Session (Session 4):

Generative AI Revolution

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